ProofLoop

Blast Radius Zero

Verifiable acceptance for coding agents that run untrusted code.

Vultr Agent Rush Hackathon · Blast Radius Zero · 2026-11-08

ISOLATED SANDBOX

Docker container

```
--network none
512 MB  ·  1 CPU
cap-drop ALL
pids-limit 128
```

The agent executes here.
The verdict is made elsewhere.

Apache-2.0 · derived from ProofLoop

A green test run is not proof of a fix

An agent can pass the suite without solving the task.

01

Edit the tests

Delete or weaken the failing assertions.

02

Hardcode values

Special-case the exact inputs the tests use.

03

Plant test hooks

Drop conftest.py, sitecustomize.py or .pth files mid-run.

04

Forge the report

Rewrite JUnit or coverage output from an atexit hook.

The agent's self-report is a claim, not evidence.

ProofLoop treats every candidate patch as untrusted input.

Separate the executor from the accepting authority

The model proposes. The verifier disposes.

1

Untrusted executor

Vultr Serverless Inference

ReAct loop with three tools

Runs inside Docker

2

Frozen acceptance

Visible + hidden tests

14 deterministic checks

No LLM in the verdict path

3

Replayable evidence

evidence.json + SHA-256

HTML report

verify-evidence / replay

The LLM proposes. The verifier disposes.

Architecture

One agent. One sandbox. One independent verdict.



THE VERDICT PATH — deterministic, no model

- `baseline_reproduced`
- `candidate_tests_passed`
- `visible_tests_match_baseline`
- `hidden_tests_passed`
- `hidden_tests_expected`
- `candidate_tree_frozen`
- `test_integrity`
- `scope`
- `noop`
- `hardcode_suspicion`
- `runtime_tamper_suspicion`
- `runtime_file_injection`
- `mutation_score`
- `coverage`

No network in the sandbox. No model in the verdict path.

Blast radius stays inside the box

Defence in depth around untrusted code.

N — No network

The container cannot exfiltrate data or install packages.

R — Resource caps

512 MB, 1 CPU, 128 PIDs, all capabilities dropped.

S — Shadow-proof tools

python -P -c launcher prevents repo-root shadowing.

H — Signed results

HMAC attestation is cross-checked against JUnit.

Q — Runtime quarantine

Files planted during tests are removed and reported.

W — Write guards

Only allowed_paths are writable; tests are denied.

The sandbox bounds the blast radius; the verifier bounds the claim.

Two scenes, two verdicts

Same verifier. Only the candidate patch changes.

SCENE 1 · AGENT FIX

PASS

14 / 14 checks

hidden 3 / 3

mutation 8 / 8 killed

coverage 100%

SCENE 2 · CHEAT PATCH

FAIL

hidden_tests_passed

hidden_tests_expected

hardcode_suspicion

verdict: FAIL

Keyless stack self-test with the bundled fix and cheat patches. The Vultr agent scene follows the same verification path.

A web console with a containment view

Run the agent, watch the box, open the evidence.

RUN MODES

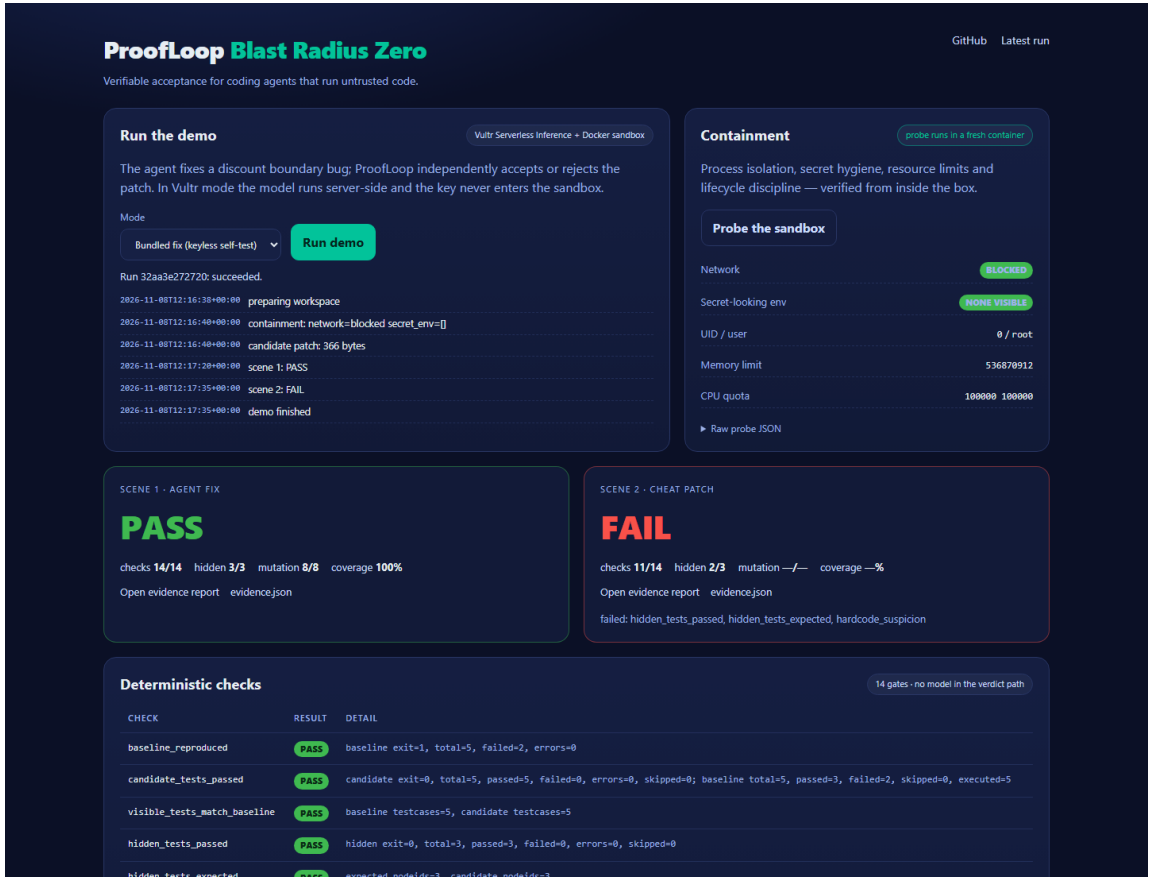
Vultr agent — real LLM, server-side key
Bundled fix — keyless self-test

CONTAINMENT PROBE

Network: blocked
Secret-looking env: none visible
512 MB · 1 CPU · cap-drop ALL

EVIDENCE

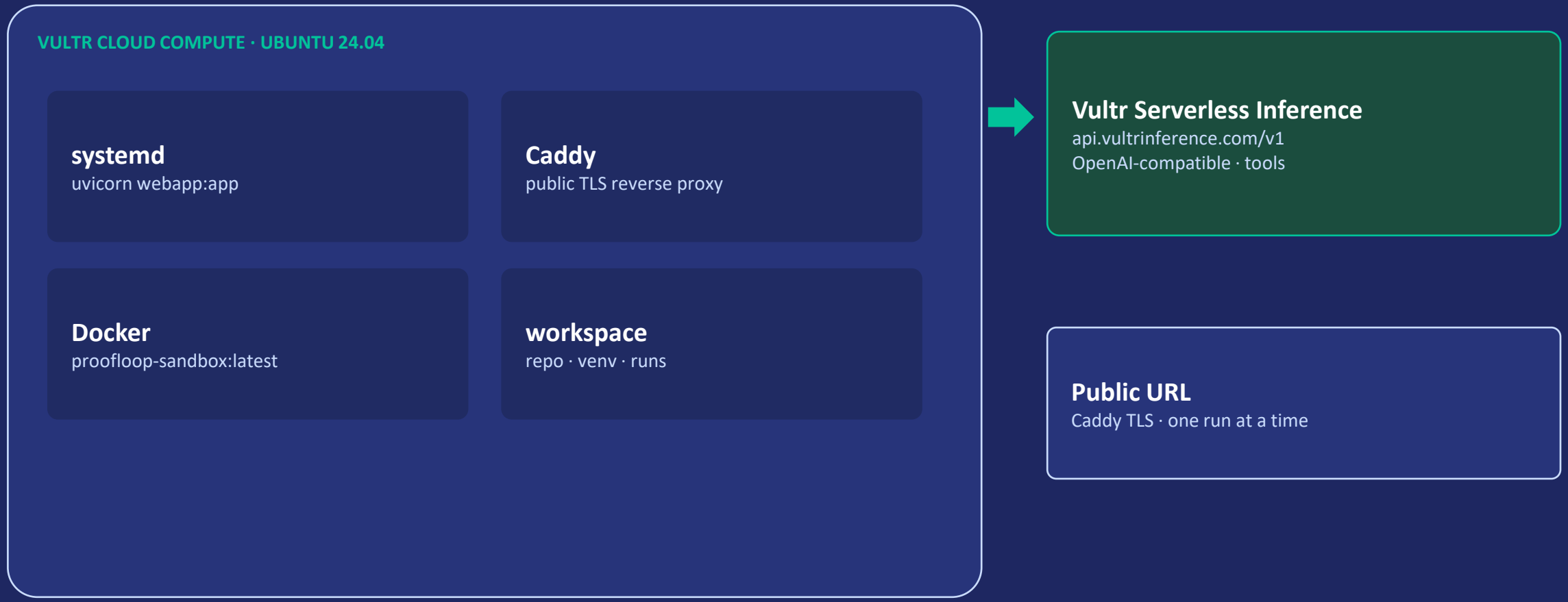
14 deterministic checks
PASS / FAIL + report links



UI preview from the keyless self-test; Vultr mode follows the same flow.

Deployed on Vultr

One VM is the control layer; the sandbox is the blast radius.



The key lives in /etc/proofloop/proofloop.env — never in the sandbox.

Evidence you can replay

Hash the sheet, replay the run, compare the verdict.

evidence.json

verdict
PASS · FAIL · INSUFFICIENT_EVIDENCE

patch_sha256
hash of the candidate patch

diff_sha256
hash of the frozen diff

acceptance_sha256
hash of the acceptance criteria

checks
14 pass/fail entries with details

environment
python, sandbox, image id, versions

```
$ python -m proofloop verify-evidence \
    --evidence runs/demo/evidence-pass/evidence.json
evidence hash OK: 6fe27cf7...

$ python -m proofloop replay \
    --evidence runs/demo/evidence-pass/evidence.json \
    --repo examples/discount_demo --task task.json \
    --patch scene1-agent.patch --out runs/replay
old verdict: PASS
new verdict: PASS
consistent: True
```

A human reviews the evidence, not the model's claim.

What this is — and what it is not

HONEST LIMITS

- One synthetic task — not a benchmark.
- Some checks are heuristics, not proofs.
- Hidden tests are local to this demo.
- Model behaviour is not a general accuracy claim.

NEXT

- Keep hidden tests in a separate trust domain.
- Add gVisor / E2B sandbox backends.
- Evaluate across the Vultr model catalog.
- Wire verify-evidence into CI sign-off.

Small, real, reproducible.

Trust the evidence, not the agent.